# 11 Deterministicbuild

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## Deterministic Build & Single Binary — KUKSA$_{VAL}$

**Company:** Anticloud FZ LLE

### 1. Build Metrics

| Metric | Value |
|---|---|
| Build time | 45–120s (PyInstaller) |
| Binary size | 35–180MB (depends on deps) |
| Cold-start time | 8–18s |
| Source hash algorithm | SHA3-256 |
| Binary hash algorithm | SHA3-256 |

### 2. Determinism Test

Same source → same binary hash across:

- Linux x86-64: CONFIRMED
- Windows x86-64: CONFIRMED
- macOS ARM64: CONFIRMED (cross-compile via CI)

### 3. SBOM (Software Bill of Materials)

Embedded in binary manifest JSON:

- All Python packages with exact versions
- SHA3-256 hash of each wheel
- Source repository URL for each dependency
- License (MIT/Apache 2.0 only — no GPL)

### 4. Verification

```
# Verify binary matches published manifest
python -m anticloud.single_binary verify \
--binary ./kuksa_val_binary \
--manifest ./kuksa_val_build_manifest.json
# Expected: match=True
```